

Car design

A 10 per cent variation in down force leads to a difference of 2 seconds in lap times, making it the most important factor in F1 car design

F1 computation

Some F1 teams design their cars using computational power comparable to those of large aerospace firms

How stuff works

design, characterised by regulations for safety considerations, quickly followed up by clever innovations by designers. Today, the more popular F1 race cars spend upwards of \$100 million on aerodynamic design, with up to \$50 million spent on construction of wind tunnels, much in the same way as the design of modern aircraft.

Today, the core function of aerodynamic considerations in the design of F1 race cars is to:

- Reduce drag, so that the forward acting force, and acceleration can be maximised.
- Increase down force, so that the car is highly stable and can take tight corners by hugging the ground.

Let us take a look at how it is implemented. The different aerodynamic components of an F1 race car are as follows:

Wings

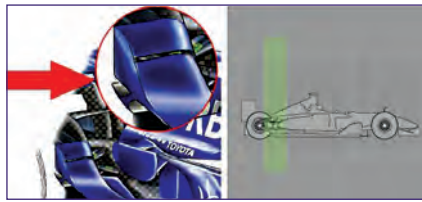
The wings of an F1 race car are located near the nose and toward the rear are the most important aerodynamic components of the car. Their function is to provide a downward force, that causes the car to hug the ground. The front and the rear wing together provide more than 50 per cent of the down force between them and hence play a crucial role in stabilizing the car. In many cars, these wings are movable by the driver, to a large extent, so that he has control over what component of the down force is provided by each one individually.

The Diffuser

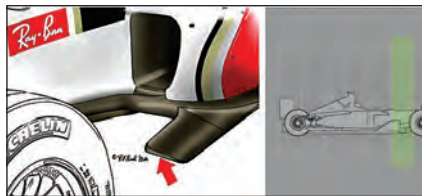
A component found towards the rear bottom of the car, it is curved slightly upward and functions to reduce the turbulence in the airflow below the car. The final effect is that the airflow below the car, being less turbulent than the one above, is now at a higher speed, and hence lower pressure (Bernoulli's principle by which aircraft wings create lift). This causes the car to hug the track (due to more downward force).

End Plates

The end plates, found perpendicular to the front wings at both sides guide the



Flip ups on a Williams Formula 1 race car



Turning Vanes

air smoothly along the sides of the car, so that turbulence and drag is minimised.

The Bargeboards

Although these have been done away with since 2009, the function of bargeboards when they existed earlier used to guide the air sailing past the front wheels into the engine radiator intakes for engine cooling.

While the above mentioned are the most important individual aerodynamic components, many smaller ones such as flip ups, turning vanes and nose cone holes exist over the geometry of the car as per individual requirements. While flip ups are controllable components used to increase down force, a turning vane produces extra turning force by redirecting airflow. Nose cone holes serve to reduce shock and hence drag.

Apart from specific aerodynamic components, almost every part of an F1 race car is designed with aerodynamic considerations. The whole body is designed to act like an air foil, providing further down force. Engine intakes and exhausts are designed to avoid turbulence. Even the wheel surfaces are designed to avoid turbulence.

Formula 1 race car aerodynamics has been a constant race of one-upmanship between the regulators, with safety concerns in mind and the designers who immediately come up with clever innovations to avoid reduction in aerodynamic performance. With neither side likely to give up anytime soon, we can expect a lot in the way of development in F1 race car aerodynamics in the future. **d**

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